

Question 1

Consider the flow of an isotropic incompressible Newtonian fluid.

- (a) Write down the four field equations governing the velocity and pressure fields in the fluid, namely the Navier–Stokes equations and the continuity equation.
- (b) Write down the equations that relate the stress field to the velocity field, namely the constitutive equations.

Question 2

Derive the Navier–Stokes equations.

Question 3

Distinguish between

- (a) steady flows and unsteady flows;
- (b) laminar flows and turbulent flows.

Question 4

In a certain viscous incompressible flow field with zero body forces, the velocity components are

$$u = ay - b(cy - y^2), \quad v = w = 0, \quad (1)$$

where a , b , and c are constants.

- (a) Use the Navier–Stokes equations to determine an expression for the pressure gradient in the x -direction.
- (b) Determine the combination of the constants a , b , and c , if any, for which the shearing stress σ_{yx} is zero at $y = 0$, where the velocity is zero.

Hint.

- (a) Substitute u , v , and w into the Navier–Stokes equations and solve for $\partial p / \partial x$.
- (b) Determine the shearing stress σ_{yx} at $y = 0$ in terms of a , b , and c , using the constitutive equation, and then set the shear stress equal to zero.

Answer.

$$(a) \quad \frac{\partial p}{\partial x} = 2b\mu, \quad (2)$$

and

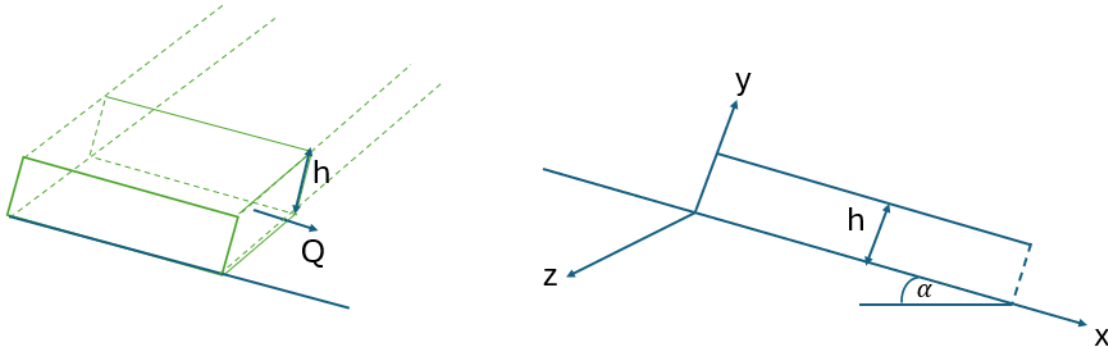
$$(b) \quad a - bc = 0. \quad (3)$$

Question 5

A thin film of viscous liquid of constant thickness, with no velocity perpendicular to the plate, flows steadily down an infinite inclined plane whose angle of inclination is α .

By means of the Navier–Stokes equations, determine the relationship between the film thickness h and the discharge per unit width Q .

The flow is laminar, and the air resistance may be neglected, so that the shearing stress at the free surface is zero.



Hint.

- (a) In the coordinates shown, determine the form of u , v , and w . The required form is

$$v = w = 0, \quad u \neq 0. \quad (4)$$

Explain why this form is appropriate.

- (b) Write down the body-force components by resolving gravity into the x -, y -, and z -directions.
- (c) Write down the boundary conditions:

$$u|_{y=0} = 0, \quad \sigma_{yx}|_{y=h} = 0, \quad p|_{y=h} = 0. \quad (5)$$

Explain the physical reason for each condition.

- (d) Write down the governing equations, namely the Navier–Stokes equations and the continuity equation, for the problem under consideration.
- (e) Solve the equations obtained in part (d), subject to the boundary conditions in part (c), to determine u .
- (f) The required discharge Q is the volume of liquid passing through a cross-section of unit width per unit time. It can be obtained by integrating $u(y)$ from 0 to h :

$$Q = \int_0^h u(y) dy. \quad (6)$$

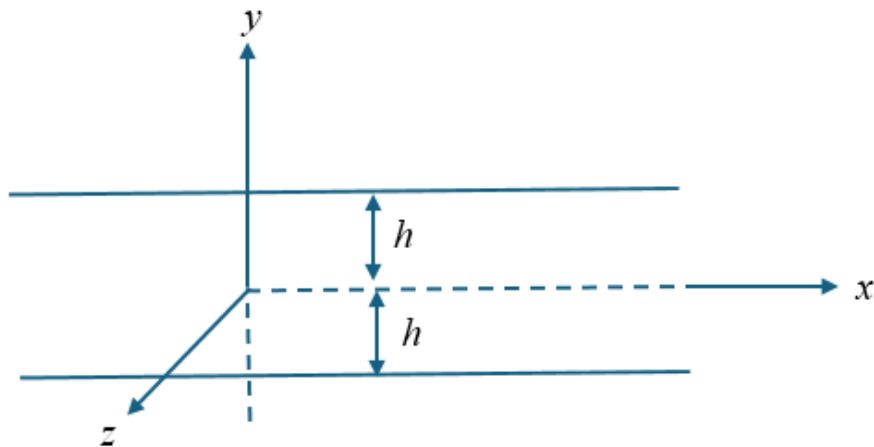
Answer.

$$Q = \frac{1}{3} \frac{\rho g \sin \alpha}{\mu} h^3 \quad (7)$$

Question 6

Consider the steady flow of an incompressible Newtonian fluid between two fixed infinite parallel plates, driven by a pressure gradient $\partial p / \partial x$.

Assume that the only body forces are due to gravity. Determine the velocity field of the flow.



Hint.

- (a) Seek a solution of the form

$$u = u(y), \quad v = w = 0. \quad (8)$$

- (b) Write down the x -, y -, and z -components of the body force in the coordinate system shown.

- (c) Write down the boundary conditions:

$$u|_{y=-h} = 0, \quad u|_{y=h} = 0. \quad (9)$$

Explain why these conditions apply.

- (d) Write down the governing equations, namely the Navier–Stokes equations and the continuity equation.
- (e) Solve the equations in part (d), subject to the boundary conditions in part (c), to determine u .

Answer.

$$u = \frac{\alpha}{2\mu} (h^2 - y^2) \quad (10)$$

where

$$\alpha = -\frac{\partial p}{\partial x} = \text{constant}. \quad (11)$$

Question 7

An incompressible Newtonian fluid is driven through the annulus

$$a \leq (x^2 + y^2)^{1/2} \leq b \quad (12)$$

in the z -direction by a uniform pressure gradient

$$\frac{dP}{dz} = -m. \quad (13)$$

Assume that the velocity field is of the form

$$v_x = v_y = 0, \quad v_z = v_z(r), \quad (14)$$

where

$$r = (x^2 + y^2)^{1/2}. \quad (15)$$

If body forces are neglected, show that v_z satisfies

$$\frac{d^2 v_z}{dr^2} + \frac{1}{r} \frac{dv_z}{dr} = -\frac{m}{\mu}, \quad (16)$$

with general solution

$$v_z = A + B \ln r - \frac{1}{4} \left(\frac{m}{\mu} \right) r^2. \quad (17)$$

If A and B are chosen to satisfy the no-slip conditions

$$v_z = 0 \quad \text{at} \quad r = a \quad \text{and} \quad r = b, \quad (18)$$

derive

$$v_z = \frac{1}{4} \left(\frac{m}{\mu} \right) \left[(a^2 - r^2) + (b^2 - a^2) \frac{\ln(r/a)}{\ln(b/a)} \right] \quad (19)$$

and deduce that the volume efflux through any section $z = \text{constant}$ is

$$Q = 2\pi \int_a^b r v_z dr. \quad (20)$$

Hence, show that

$$Q = \frac{\pi m b^4}{8\mu} \left[1 - q^4 + (1 - q^2)^2 (\ln q)^{-1} \right] \quad (21)$$

where

$$q = \frac{a}{b}. \quad (22)$$